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CC02 — Lab Report

Digital Signal Processing

Lab 6

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1 Lab 6

1.1 Overview

This is the report of lab 6.

1.2 Objectives

- Analyze systems using Z -transform, Z^- -transform, Z^+ -Transform.

1.3 Methodologies

1.3.1 Hardware & Software

- Hardware: Intel-based Laptop
- Software: Scilab (Windows OS)

2 Theory Background

Continuous-Time Periodic Signals

- **Fourier series**

- $x(t)$: continuous-time, periodic with fundamental period $T_p = \frac{1}{F_0}$
- Synthesis Equation:

$$x(t) = \sum_{k=-\infty}^{+\infty} c_k e^{j2\pi k F_0 t}$$

- Suppose that $x_k(t) = c_k e^{j2\pi k F_0 t}$
 - * $x_k(t)$ is periodic with a period $T_k = \frac{T_p}{k}$ (frequency kF_0)
 - * $x_k(t)$ contributes to $x(t)$ a value c_k
- Coefficients of Fourier series (Analysis Equation):

$$c_k = \frac{1}{T_p} \int_{T_p} x(t) e^{-j2\pi k F_0 t} dt$$

- $c_k = |c_k| e^{j\theta_k}$ (Amplitude and Phase)

Continuous-Time Aperiodic Signals

- **Fourier Transform**

- $x(t)$: continuous-time aperiodic
- Analysis Equation (Fourier Transform):

$$X(F) = \int_{-\infty}^{+\infty} x(t) e^{-j2\pi F t} dt$$

- Fourier coefficients:

$$c_k = \frac{1}{T_p} X(kF_0) = F_0 X(kF_0)$$

- Synthesis Equation (Inverse Fourier Transform):

$$x(t) = \int_{-\infty}^{+\infty} X(F) e^{j2\pi Ft} dF$$

Discrete-Time Periodic Signals

- $x(n)$ is periodic with a period $N \Rightarrow x(n+N) = x(n), \forall n$
- Fourier series has maximum N frequency components in range $[0, 2\pi]$ or $[-\pi, \pi]$
- **Discrete-time Fourier Series (DTFS)**

- Synthesis Equation:

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi \frac{k}{N} n}$$

- Analysis Equation:

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi \frac{k}{N} n}$$

- Notes:
 - Express $x(n)$ in frequency domain: c_k represents amplitude and phase.
 - $c_{k+N} = c_k$: Spectrum of $x(n)$ with period N is periodic with period N .

Discrete-Time Periodic Signals (Real Signals and Symmetry)

- If $x(n)$ is real, $x^*(n) = x(n) \Rightarrow c_k^* = c_{-k}$
- This implies:

$$\begin{cases} |c_{-k}| = |c_k| \\ \angle c_{-k} = -\angle c_k \end{cases}$$

- Besides, $c_{N+k} = c_k$, we have:

$$\begin{cases} |c_k| = |c_{N-k}| \\ \angle c_k = -\angle c_{N-k} \end{cases}$$

- Implications:
 - Amplitude spectrum is **even symmetric**
 - Phase spectrum is **odd symmetric**

Discrete-Time Aperiodic Signals

- **Fourier Transform**

- Analysis Equation:

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

– Synthesis Equation:

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

Fourier Transform and Z-Transform

$$\text{Z Transform: } X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$\text{Fourier Transform: } X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

With $z = e^{j\omega}$

Properties of Fourier Transform

- **Linearity:**

$$x_1(n) \xleftrightarrow{\mathcal{F}} X_1(\omega), \quad x_2(n) \xleftrightarrow{\mathcal{F}} X_2(\omega) \Rightarrow a_1 x_1(n) + a_2 x_2(n) \xleftrightarrow{\mathcal{F}} a_1 X_1(\omega) + a_2 X_2(\omega)$$

- **Time shifting:**

$$x(n) \xleftrightarrow{\mathcal{F}} X(\omega) \Rightarrow x(n-k) \xleftrightarrow{\mathcal{F}} e^{-j\omega k} X(\omega)$$

- **Inverse in time domain:**

$$x(n) \xleftrightarrow{\mathcal{F}} X(\omega) \Rightarrow x(-n) \xleftrightarrow{\mathcal{F}} X(-\omega)$$

- **Convolution:**

$$x_1(n) \xleftrightarrow{\mathcal{F}} X_1(\omega), \quad x_2(n) \xleftrightarrow{\mathcal{F}} X_2(\omega) \Rightarrow x(n) = x_1(n) * x_2(n) \xleftrightarrow{\mathcal{F}} X(\omega) = X_1(\omega) X_2(\omega)$$

- **Frequency shifting:**

$$x(n) \xleftrightarrow{\mathcal{F}} X(\omega) \Rightarrow e^{j\omega_0 n} x(n) \xleftrightarrow{\mathcal{F}} X(\omega - \omega_0)$$

- **Modulation:**

$$x(n) \xleftrightarrow{\mathcal{F}} X(\omega) \Rightarrow x(n) \cos(\omega_0 n) \xleftrightarrow{\mathcal{F}} \frac{1}{2} [X(\omega + \omega_0) + X(\omega - \omega_0)]$$

- **Parseval:**

$$x_1(n) \xleftrightarrow{\mathcal{F}} X_1(\omega), \quad x_2(n) \xleftrightarrow{\mathcal{F}} X_2(\omega) \Rightarrow \sum_{n=-\infty}^{+\infty} x_1(n) x_2^*(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\omega) X_2^*(\omega) d\omega$$

• **Differentiation in frequency domain:**

$$x(n) \xleftrightarrow{\mathcal{F}} X(\omega) \Rightarrow nx(n) \xleftrightarrow{\mathcal{F}} j \frac{dX(\omega)}{d\omega}$$

LTI Systems in Frequency Domain Time Domain:

$$y(n) = x(n) * h(n)$$

Frequency Domain:

$$y(n) \xleftrightarrow{\mathcal{F}} Y(\omega) = X(\omega)H(\omega)$$

$H(\omega)$ can be expressed as $H(\omega) = |H(\omega)|e^{j\theta(\omega)}$

$$H(\omega) = \sum_{k=-\infty}^{+\infty} h(k)e^{-j\omega k} = \sum_{k=-\infty}^{+\infty} h(k) \cos(\omega k) - j \sum_{k=-\infty}^{+\infty} h(k) \sin(\omega k)$$

$$H(\omega) = H_R(\omega) + jH_I(\omega) = \sqrt{H_R^2(\omega) + H_I^2(\omega)} e^{j \tan^{-1} \left(\frac{H_I(\omega)}{H_R(\omega)} \right)}$$

Where:

$$H_R(\omega) = \sum_{k=-\infty}^{+\infty} h(k) \cos(\omega k) \quad (\text{Even Function})$$

$$H_I(\omega) = - \sum_{k=-\infty}^{+\infty} h(k) \sin(\omega k) \quad (\text{Odd Function})$$

Therefore:

$$|H(\omega)| = \sqrt{H_R^2(\omega) + H_I^2(\omega)} \quad (\text{Even})$$

$$\Theta(\omega) = e^{\tan^{-1} \left(\frac{H_I(\omega)}{H_R(\omega)} \right)} \quad (\text{Odd})$$

Periodic input signal:

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j \frac{2\pi k}{N} n}, \quad y(n) = \sum_{k=0}^{N-1} c_k H \left(\frac{2\pi k}{N} \right) e^{j \frac{2\pi k}{N} n}$$

Response to periodic input is also periodic with same period N .

Aperiodic input signal:

$$x(n) \xrightarrow{\mathcal{F}} X(\omega), \quad h(n) \xrightarrow{\mathcal{F}} H(\omega), \quad y(n) = x(n) * h(n) \xrightarrow{\mathcal{F}} Y(\omega) = X(\omega)H(\omega)$$

At specific frequency ω_0 :

$$Y(\omega_0) = X(\omega_0)H(\omega_0) = |H(\omega_0)|e^{j\Theta(\omega_0)}X(\omega_0)$$



Amplitude: scaling $|H(\omega_0)|$, Phase: shift $\Theta(\omega_0)$

System function $H(z)$ vs. Frequency response function $H(\omega)$

$$H(\omega) = H(z) \Big|_{z=e^{j\omega}} = \sum_{n=-\infty}^{+\infty} h(n)e^{-j\omega n}$$

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} \Rightarrow H(\omega) = \frac{\sum_{k=0}^M b_k e^{-j\omega k}}{1 + \sum_{k=1}^N a_k e^{-j\omega k}}$$

$$|H(\omega)|^2 = H(\omega)H^*(\omega) = H(\omega)H(-\omega) = H(z)H(z^{-1})$$

3 Exercise

3.1 Exercise 1

Part a: Fourier Transform of a Triangular Pulse

Given the signal:

$$x_1(t) = \begin{cases} 1 - \frac{|t|}{\tau}, & |t| \leq \tau \\ 0, & |t| > \tau \end{cases}$$

By the definition of the continuous-time Fourier transform:

$$X_1(\omega) = \int_{-\infty}^{\infty} x_1(t) e^{-j\omega t} dt = \int_{-\tau}^{\tau} \left(1 - \frac{|t|}{\tau}\right) e^{-j\omega t} dt$$

Using Euler's identity ($e^{-j\omega t} = \cos(\omega t) - j \sin(\omega t)$), we can separate the integral into real and imaginary parts. Because $x_1(t)$ is an even function and $\sin(\omega t)$ is an odd function, their product is odd, and the integral of the sine term over the symmetric interval $[-\tau, \tau]$ is zero. We are left with the even cosine term, allowing us to change the lower limit to 0 and double the result:

$$X_1(\omega) = 2 \int_0^{\tau} \left(1 - \frac{t}{\tau}\right) \cos(\omega t) dt$$

We evaluate this using integration by parts ($\int u dv = uv - \int v du$). Let:

$$u = 1 - \frac{t}{\tau} \implies du = -\frac{1}{\tau} dt$$

$$dv = \cos(\omega t) dt \implies v = \frac{\sin(\omega t)}{\omega}$$

Applying the formula:

$$X_1(\omega) = 2 \left[\left(1 - \frac{t}{\tau}\right) \frac{\sin(\omega t)}{\omega} \right]_0^{\tau} - \int_0^{\tau} \frac{\sin(\omega t)}{\omega} \left(-\frac{1}{\tau}\right) dt$$

Evaluating the boundary term at $t = \tau$ and $t = 0$ yields 0. The remaining integral is:

$$X_1(\omega) = \frac{2}{\omega\tau} \int_0^{\tau} \sin(\omega t) dt = \frac{2}{\omega\tau} \left[-\frac{\cos(\omega t)}{\omega} \right]_0^{\tau}$$

Evaluating from 0 to τ :

$$X_1(\omega) = \frac{2}{\omega^2\tau} (-\cos(\omega\tau) - (-\cos(0))) = \frac{2}{\omega^2\tau} (1 - \cos(\omega\tau))$$

Using the half-angle trigonometric identity $1 - \cos(\theta) = 2 \sin^2\left(\frac{\theta}{2}\right)$ with $\theta = \omega\tau$:

$$X_1(\omega) = \frac{2}{\omega^2\tau} \left[2 \sin^2\left(\frac{\omega\tau}{2}\right) \right] = \frac{4 \sin^2\left(\frac{\omega\tau}{2}\right)}{\omega^2\tau}$$

To format this into the standard sampling function $\text{Sa}(x) = \frac{\sin x}{x}$, we multiply the numerator and denominator by τ :

$$X_1(\omega) = \tau \frac{\sin^2\left(\frac{\omega\tau}{2}\right)}{\frac{\omega^2\tau^2}{4}} = \tau \left[\frac{\sin\left(\frac{\omega\tau}{2}\right)}{\frac{\omega\tau}{2}} \right]^2$$

Final Result for Part a:

$$X_1(\omega) = \tau \text{Sa}^2\left(\frac{\omega\tau}{2}\right)$$

Part b: Fourier Transform of a Complex Exponential

Given the periodic signal:

$$x_2(t) = e^{j\omega_0 t}$$

Because this is a periodic signal with infinite energy, it does not satisfy the Dirichlet conditions for absolute integrability. Thus, its Fourier transform must be represented using the Dirac delta function (δ).

We start with the known Fourier transform pair of a constant DC signal:

$$1 \longleftrightarrow 2\pi\delta(\omega)$$

Next, we apply the frequency shifting property of the Fourier transform, which states that multiplying a time-domain signal by a complex exponential shifts its frequency spectrum:

$$x(t)e^{j\omega_0 t} \longleftrightarrow X(\omega - \omega_0)$$

Applying this shift to our constant signal:

$$1 \cdot e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)$$

Final Result for Part b:

$$X_2(\omega) = 2\pi\delta(\omega - \omega_0)$$

3.2 Exercise 2

Part a: $x_1(n) = u(n) - u(n - 6)$

This signal is a rectangular pulse in discrete time, existing only for $n = 0, 1, 2, 3, 4, 5$. By the definition of the Discrete-Time Fourier Transform (DTFT):

$$X_1(\omega) = \sum_{n=-\infty}^{\infty} x_1(n)e^{-j\omega n} = \sum_{n=0}^5 (1)e^{-j\omega n}$$

Using the finite geometric series formula $\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r}$ with $r = e^{-j\omega}$:

$$X_1(\omega) = \frac{1 - e^{-j6\omega}}{1 - e^{-j\omega}}$$

To express this in the standard magnitude and phase format (Dirichlet kernel), we factor out half the angle from both the numerator and denominator:

$$X_1(\omega) = \frac{e^{-j3\omega}(e^{j3\omega} - e^{-j3\omega})}{e^{-j\omega/2}(e^{j\omega/2} - e^{-j\omega/2})}$$

Using Euler's identity ($2j \sin(\theta) = e^{j\theta} - e^{-j\theta}$):

$$X_1(\omega) = \frac{e^{-j3\omega} \cdot 2j \sin(3\omega)}{e^{-j\omega/2} \cdot 2j \sin(\omega/2)} = e^{-j2.5\omega} \frac{\sin(3\omega)}{\sin(\omega/2)}$$

Part b: $x_2(n) = 2^n u(-n)$

This signal exists only for $n \leq 0$. Applying the DTFT:

$$X_2(\omega) = \sum_{n=-\infty}^0 2^n e^{-j\omega n}$$

Let $m = -n$. As n goes from 0 to $-\infty$, m goes from 0 to ∞ :

$$X_2(\omega) = \sum_{m=0}^{\infty} 2^{-m} e^{j\omega m} = \sum_{m=0}^{\infty} \left(\frac{1}{2} e^{j\omega}\right)^m$$

Since the magnitude $\left|\frac{1}{2} e^{j\omega}\right| = 0.5 < 1$, the infinite geometric series converges using $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$:

$$X_2(\omega) = \frac{1}{1 - \frac{1}{2} e^{j\omega}}$$

Part c: $x_3(n) = \left(\frac{1}{4}\right)^n u(n+4)$

This is a decaying exponential that starts at $n = -4$. Applying the DTFT:

$$X_3(\omega) = \sum_{n=-4}^{\infty} \left(\frac{1}{4}\right)^n e^{-j\omega n}$$

To evaluate this, apply a change of variables to shift the index to start at 0. Let $m = n + 4$, which means $n = m - 4$:

$$X_3(\omega) = \sum_{m=0}^{\infty} \left(\frac{1}{4}\right)^{m-4} e^{-j\omega(m-4)}$$

Factor out the constant terms (those that do not depend on m):

$$X_3(\omega) = \left(\frac{1}{4}\right)^{-4} e^{j4\omega} \sum_{m=0}^{\infty} \left(\frac{1}{4}\right)^m e^{-j\omega m} = 256e^{j4\omega} \sum_{m=0}^{\infty} \left(\frac{1}{4}e^{-j\omega}\right)^m$$

Applying the infinite geometric series formula:

$$X_3(\omega) = \frac{256e^{j4\omega}}{1 - \frac{1}{4}e^{-j\omega}}$$

Part d: $x_4(n) = 2 - \frac{1}{2}|n|$ for $|n| \leq 4$

This is an even, discrete-time triangular pulse. First, evaluate the non-zero samples:

$$\begin{aligned}x_4(0) &= 2 \\x_4(1) &= x_4(-1) = 1.5 \\x_4(2) &= x_4(-2) = 1 \\x_4(3) &= x_4(-3) = 0.5 \\x_4(4) &= x_4(-4) = 0\end{aligned}$$

Because the signal is an even function ($x(n) = x(-n)$), the sine terms in the Fourier transform cancel out, leaving only the real cosine terms. We group the symmetric points:

$$X_4(\omega) = x_4(0) + \sum_{n=1}^3 x_4(n) (e^{-j\omega n} + e^{j\omega n})$$

Using Euler's identity ($e^{j\theta} + e^{-j\theta} = 2\cos\theta$):

$$X_4(\omega) = x_4(0) + 2 \sum_{n=1}^3 x_4(n) \cos(\omega n)$$

Substitute the values:

$$X_4(\omega) = 2 + 2[1.5 \cos(\omega) + 1 \cos(2\omega) + 0.5 \cos(3\omega)]$$

$$X_4(\omega) = 2 + 3 \cos(\omega) + 2 \cos(2\omega) + \cos(3\omega)$$

Part e: $x_5(n) = |a|^n \sin(\omega_0 n)$ **with** $|a| < 1$

Note: For the sum to converge, this signal is assumed to be causal, implied as $n \geq 0$, otherwise the exponential term blows up as $n \rightarrow -\infty$. We proceed by evaluating from $n = 0$ to ∞ .

Using Euler's identity for the sine function: $\sin(\omega_0 n) = \frac{e^{j\omega_0 n} - e^{-j\omega_0 n}}{2j}$

$$X_5(\omega) = \sum_{n=0}^{\infty} |a|^n \left(\frac{e^{j\omega_0 n} - e^{-j\omega_0 n}}{2j} \right) e^{-j\omega n}$$

Split the sum into two separate geometric series:

$$X_5(\omega) = \frac{1}{2j} \left[\sum_{n=0}^{\infty} \left(|a| e^{j(\omega_0 - \omega)} \right)^n - \sum_{n=0}^{\infty} \left(|a| e^{-j(\omega_0 + \omega)} \right)^n \right]$$

Apply the geometric series sum formula to both terms:

$$X_5(\omega) = \frac{1}{2j} \left[\frac{1}{1 - |a| e^{j\omega_0} e^{-j\omega}} - \frac{1}{1 - |a| e^{-j\omega_0} e^{-j\omega}} \right]$$

Find a common denominator to combine the fractions:

$$X_5(\omega) = \frac{1}{2j} \left[\frac{(1 - |a| e^{-j\omega_0} e^{-j\omega}) - (1 - |a| e^{j\omega_0} e^{-j\omega})}{(1 - |a| e^{j\omega_0} e^{-j\omega})(1 - |a| e^{-j\omega_0} e^{-j\omega})} \right]$$

Simplify the numerator:

$$\text{Numerator} = |a| e^{-j\omega} (e^{j\omega_0} - e^{-j\omega_0}) = |a| e^{-j\omega} (2j \sin \omega_0)$$

Expand the denominator:

$$\text{Denominator} = 1 - |a| e^{-j\omega} (e^{j\omega_0} + e^{-j\omega_0}) + |a|^2 e^{-j2\omega} = 1 - 2|a| \cos(\omega_0) e^{-j\omega} + |a|^2 e^{-j2\omega}$$

Recombining and cancelling the $2j$ term leaves the standard formula:

$$X_5(\omega) = \frac{|a| \sin(\omega_0) e^{-j\omega}}{1 - 2|a| \cos(\omega_0) e^{-j\omega} + |a|^2 e^{-j2\omega}}$$

3.3 Exercise 3

The discrete-time signals analyzed in this exercise are defined as follows:

$$x_1(n) = 0.1^n u(n)$$

$$x_2(n) = \delta(n) + \delta(n-1) + \delta(n-2) + \delta(n-3)$$

The corresponding amplitude and phase spectra for both signals were plotted using Scilab and are presented in Figure 1 and Figure 2.

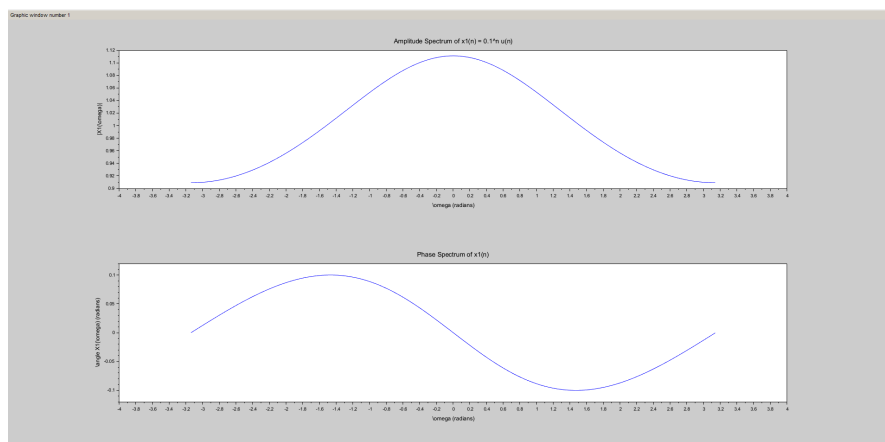


Figure 1: Amplitude and phase spectrum for $x_1(n)$.

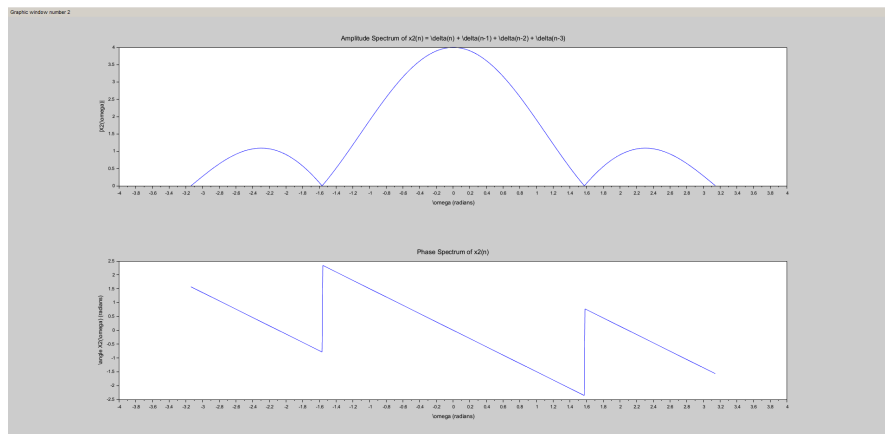


Figure 2: Amplitude and phase spectrum for $x_2(n)$.

3.4 Exercise 4

Alternative Derivation using Complex Conjugate (Exercise 4)

From the difference equation, the frequency response is:

$$H(\omega) = \frac{1}{1 + 0.1e^{-j\omega} + 0.2e^{-j2\omega}}$$

To separate the real and imaginary parts, we multiply the numerator and denominator by the complex conjugate of the denominator:

$$H(\omega) = \frac{1}{1 + 0.1e^{-j\omega} + 0.2e^{-j2\omega}} \cdot \frac{1 + 0.1e^{j\omega} + 0.2e^{j2\omega}}{1 + 0.1e^{j\omega} + 0.2e^{j2\omega}}$$

Expanding the denominator yields:

$$\begin{aligned} D(\omega) &= (1 + 0.1e^{-j\omega} + 0.2e^{-j2\omega})(1 + 0.1e^{j\omega} + 0.2e^{j2\omega}) \\ &= 1(1 + 0.1e^{j\omega} + 0.2e^{j2\omega}) + 0.1e^{-j\omega}(1 + 0.1e^{j\omega} + 0.2e^{j2\omega}) + 0.2e^{-j2\omega}(1 + 0.1e^{j\omega} + 0.2e^{j2\omega}) \\ &= 1 + 0.1e^{j\omega} + 0.2e^{j2\omega} + 0.1e^{-j\omega} + 0.01 + 0.02e^{j\omega} + 0.2e^{-j2\omega} + 0.02e^{-j\omega} + 0.04 \end{aligned}$$

Grouping the constant terms and conjugate pairs:

$$D(\omega) = (1 + 0.01 + 0.04) + 0.12(e^{j\omega} + e^{-j\omega}) + 0.2(e^{j2\omega} + e^{-j2\omega})$$

Using the identity $e^{j\theta} + e^{-j\theta} = 2 \cos \theta$, the denominator becomes a purely real function:

$$D(\omega) = 1.05 + 0.24 \cos(\omega) + 0.4 \cos(2\omega)$$

Expanding the numerator using Euler's formula ($e^{j\theta} = \cos \theta + j \sin \theta$):

$$N(\omega) = (1 + 0.1 \cos \omega + 0.2 \cos 2\omega) + j(0.1 \sin \omega + 0.2 \sin 2\omega)$$

We can now define the real part $H_R(\omega)$ and imaginary part $H_I(\omega)$:

$$H_R(\omega) = \frac{1 + 0.1 \cos \omega + 0.2 \cos 2\omega}{1.05 + 0.24 \cos \omega + 0.4 \cos 2\omega}$$

$$H_I(\omega) = \frac{0.1 \sin \omega + 0.2 \sin 2\omega}{1.05 + 0.24 \cos \omega + 0.4 \cos 2\omega}$$

The amplitude and phase spectra are then given by:

$$|H(\omega)| = \sqrt{(H_R(\omega))^2 + (H_I(\omega))^2}$$

$$\angle H(\omega) = \tan^{-1} \left(\frac{H_I(\omega)}{H_R(\omega)} \right)$$

3.5 Exercise 5

Problem Statement:

Given an LTI system with impulse response $h(n) = \delta(n) + \delta(n - 1)$. Determine the output $y(n)$ when the input is $x(n) = 0.5^n u(n)$ using the Fourier Transform. Then, plot the amplitude and phase spectrums using Scilab.

Mathematical Derivation:

Step 1: Find the DTFT of $h(n)$ and $x(n)$

First, we find the Discrete-Time Fourier Transform (DTFT) of the impulse response. Using the time-shifting property $\delta(n - k) \longleftrightarrow e^{-j\omega k}$, we get:

$$H(\omega) = 1 + e^{-j\omega}$$

Next, we find the DTFT of the input signal. Using the standard transform pair for a causal exponential decay $a^n u(n) \longleftrightarrow \frac{1}{1 - ae^{-j\omega}}$, we get:

$$X(\omega) = \frac{1}{1 - 0.5e^{-j\omega}}$$

Step 2: Calculate the Output in the Frequency Domain

For an LTI system, convolution in the time domain corresponds to multiplication in the frequency domain. Therefore, the output spectrum $Y(\omega)$ is:

$$Y(\omega) = H(\omega)X(\omega) = (1 + e^{-j\omega}) \left(\frac{1}{1 - 0.5e^{-j\omega}} \right)$$
$$Y(\omega) = \frac{1 + e^{-j\omega}}{1 - 0.5e^{-j\omega}}$$

Step 3: Apply the Inverse DTFT to find $y(n)$

To simplify the inverse transform, we can separate the numerator into two distinct terms:

$$Y(\omega) = \frac{1}{1 - 0.5e^{-j\omega}} + \frac{e^{-j\omega}}{1 - 0.5e^{-j\omega}}$$

Now, we apply the inverse DTFT. The first term is exactly the transform of our original input signal $x(n)$. The second term is multiplied by $e^{-j\omega}$, which, according to the time-shifting property, represents a delay of 1 sample in the time domain:

$$y(n) = x(n) + x(n - 1)$$

Substituting the definition of $x(n)$ back into the equation yields the final time-domain output:

$$y(n) = 0.5^n u(n) + 0.5^{n-1} u(n-1)$$

Scilab Visualization:

The frequency response of the output signal $Y(\omega)$ was computed and plotted using Scilab. The amplitude spectrum $|Y(\omega)|$ and the phase spectrum $\angle Y(\omega)$ are shown below in Figure 3.

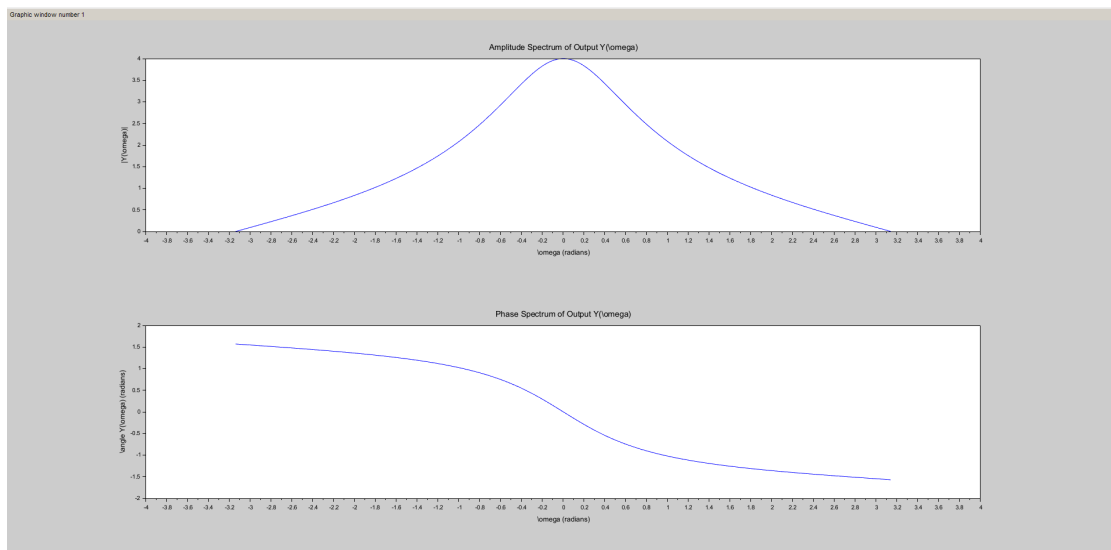


Figure 3: Amplitude and phase spectrums of the output signal $Y(\omega)$ for Exercise 5.

4 Additional Exercise

4.1 Exercise 4.1

Solution

(a)

$x_a(t)$ is periodic, its Fourier Series:

$$\begin{aligned}x_a(t) &= \sum_{k=-\infty}^{\infty} c_k e^{j2\pi kt/\tau} \\c_k &= \frac{1}{\tau} \int_0^{\tau} A \sin(\pi t/\tau) e^{-j2\pi kt/\tau} dt \\&= \frac{A}{j2\tau} \int_0^{\tau} \left[e^{j\pi t/\tau} - e^{-j\pi t/\tau} \right] e^{-j2\pi kt/\tau} dt \\&= \frac{A}{j2\tau} \left[\frac{e^{j\pi(1-2k)t/\tau}}{j\frac{\pi}{\tau}(1-2k)} - \frac{e^{-j\pi(1+2k)t/\tau}}{-j\frac{\pi}{\tau}(1+2k)} \right]_0^{\tau} \\&= \frac{A}{\pi} \left(\frac{1}{1-2k} + \frac{1}{1+2k} \right) \\&= \frac{2A}{\pi(1-4k^2)}\end{aligned}$$

$$\begin{aligned}X_a(F) &= \int_{-\infty}^{\infty} x_a(t) e^{-j2\pi Ft} dt \\&= \sum_{k=-\infty}^{\infty} c_k \int_{-\infty}^{\infty} e^{-j2\pi(F-k/\tau)t} dt \\&= \sum_{k=-\infty}^{\infty} c_k \delta\left(F - \frac{k}{\tau}\right)\end{aligned}$$

The spectrum of $x_a(t)$ consists of spectral lines of frequencies k/τ , $k = 0, \pm 1, \pm 2, \dots$ with amplitude $|c_k|$ and phases $\angle c_k$.

(b)

$$P_x = \frac{1}{\tau} \int_0^{\tau} x_a^2(t) dt = \frac{1}{\tau} \int_0^{\tau} A^2 \sin^2\left(\frac{\pi t}{\tau}\right) dt = \frac{A^2}{2}$$

(c)

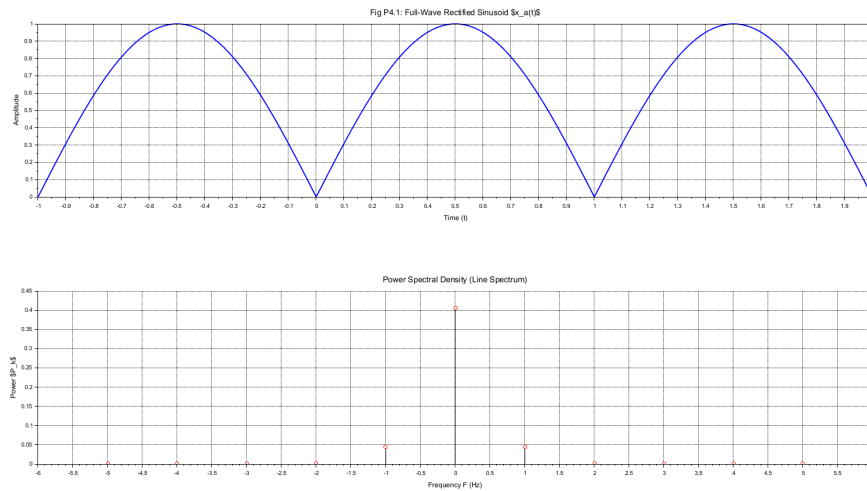


Figure 4: *Caption*

(d)

$$\begin{aligned}
 P_x &= \frac{1}{\tau} \int_0^\tau x_a^2(t) dt \\
 &= \sum |c_k|^2 \\
 \sum_{k=-\infty}^{\infty} |c_k|^2 &= \frac{4A^2}{\pi^2} \sum_{k=-\infty}^{\infty} \frac{1}{(4k^2 - 1)^2} = \frac{4A^2}{\pi^2} \left[1 + \frac{2}{3^2} + \frac{2}{15^2} + \dots \right]
 \end{aligned}$$

with $1 + \frac{2}{3^2} + \frac{2}{15^2} + \dots = 1.2337$ (Infinite series sum to $\frac{\pi^2}{8}$)

$$\Rightarrow \sum_{k=-\infty}^{\infty} |c_k|^2 = \frac{4A^2}{\pi^2} (1.2337) = \frac{A^2}{2}$$

4.2 Exercise 4.2

Solution

(a)

$$\begin{aligned}
 x_a(t) &= Ae^{-at}u(t), \quad a > 0 \\
 X_a(F) &= \int_0^\infty Ae^{-at}e^{-j2\pi Ft} dt \\
 &= \left[\frac{A}{-a - j2\pi F} e^{-(a+j2\pi F)t} \right]_0^\infty \\
 &= \frac{A}{a + j2\pi F}
 \end{aligned}$$

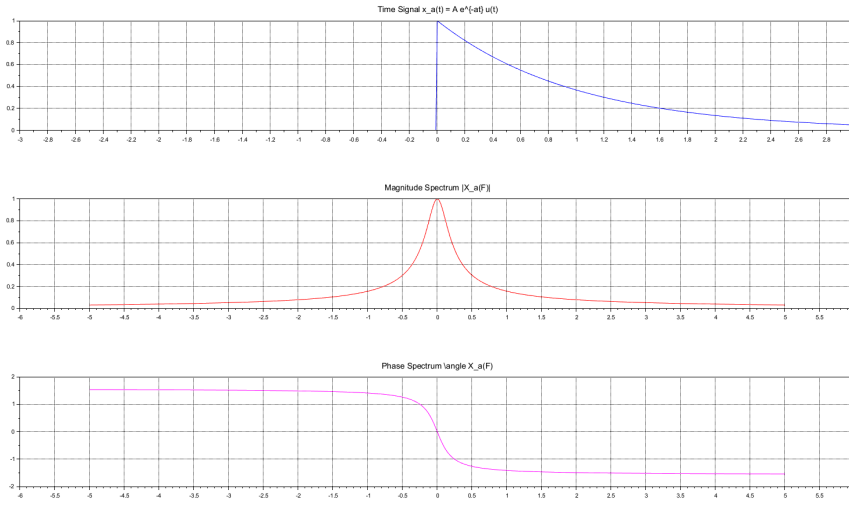


Figure 5: *Caption*

$$|X_a(F)| = \frac{A}{\sqrt{a^2 + (2\pi F)^2}}$$

$$\angle X_a(F) = -\tan^{-1} \left(\frac{2\pi F}{a} \right)$$

(b)

$$\begin{aligned} X_a(F) &= \int_{-\infty}^0 A e^{at} e^{-j2\pi Ft} dt + \int_0^{\infty} A e^{-at} e^{-j2\pi Ft} dt \\ &= \frac{A}{a - j2\pi F} + \frac{A}{a + j2\pi F} \\ &= \frac{2aA}{a^2 + (2\pi F)^2} \end{aligned}$$

$$|X_a(F)| = \frac{2aA}{a^2 + (2\pi F)^2}$$

$$\angle X_a(F) = 0$$

4.3 Exercise 4.3

Solution

(a)

$$X_a(F) = \int_{-\tau}^0 \left(1 + \frac{t}{\tau} \right) e^{-j2\pi Ft} dt + \int_0^{\tau} \left(1 - \frac{t}{\tau} \right) e^{-j2\pi Ft} dt$$

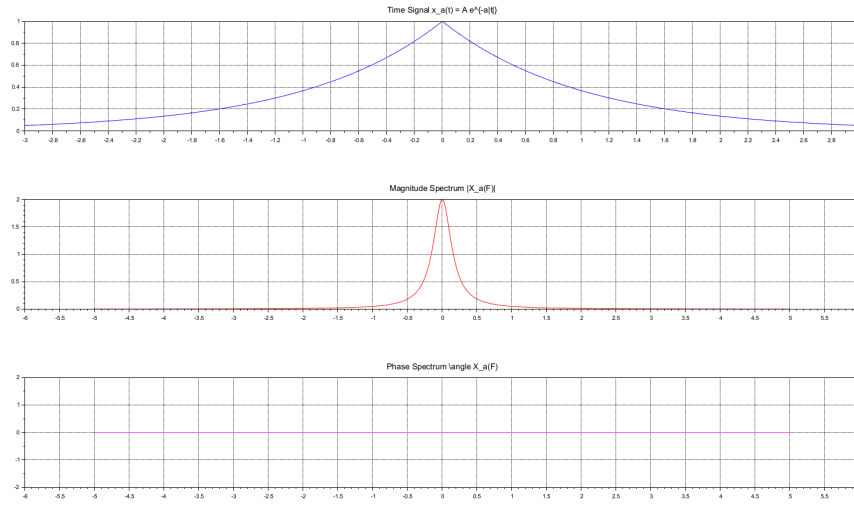


Figure 6: *Caption*

Or we can calculate using:

$$y(t) = x'(t) = \begin{cases} \frac{1}{\tau}, & -\tau < t \leq 0 \\ -\frac{1}{\tau}, & 0 < t \leq \tau \end{cases}$$

$$\begin{aligned} Y(F) &= \int_{-\tau}^{\tau} y(t) e^{-j2\pi Ft} dt \\ &= \int_{-\tau}^0 \frac{1}{\tau} e^{-j2\pi Ft} dt + \int_0^{\tau} \left(-\frac{1}{\tau}\right) e^{-j2\pi Ft} dt \\ &= \frac{-2 \sin^2(\pi F \tau)}{j\pi F \tau} \end{aligned}$$

$$\begin{aligned} X(F) &= \frac{1}{j2\pi F} Y(F) \\ &= \tau \left(\frac{\sin(\pi F \tau)}{\pi F \tau} \right)^2 \end{aligned}$$

$$\begin{aligned} |X(F)| &= \tau \left(\frac{\sin(\pi F \tau)}{\pi F \tau} \right)^2 \\ \angle X_a(F) &= 0 \end{aligned}$$

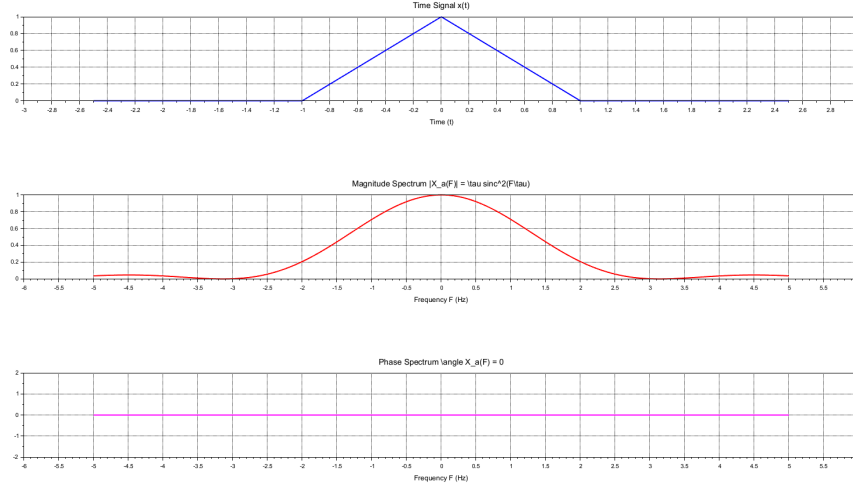


Figure 7: *Caption*

(b)

$$\begin{aligned}
 c_k &= \frac{1}{T_p} \int_{-T_p/2}^{T_p/2} x(t) e^{-j2\pi kt/T_p} dt \\
 &= \frac{1}{T_p} \left[\int_{-\tau}^0 \left(1 + \frac{t}{\tau}\right) e^{-j2\pi kt/T_p} dt + \int_0^{\tau} \left(1 - \frac{t}{\tau}\right) e^{-j2\pi kt/T_p} dt \right] \\
 &= \frac{\tau}{T_p} \left[\frac{\sin(\pi k \tau / T_p)}{\pi k \tau / T_p} \right]^2
 \end{aligned}$$

(c)

From (a) and (b) $\Rightarrow c_k = \frac{1}{T_p} X_a\left(\frac{k}{T_p}\right)$

4.4 Exercise 4.4

Solution

(a) Signal Sketch and Spectra

The given periodic sequence is $x(n) = \{\dots, 1, 0, 1, 2, 3, 2, 1, 0, 1, \dots\}$. By observation, the signal repeats every $N = 6$ samples. One fundamental period is defined as:

$$x_N(n) = \{3, 2, 1, 0, 1, 2\} \quad \text{for } n = 0, 1, \dots, 5$$

The Discrete Fourier Series (DFS) coefficients are calculated as:

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn}$$

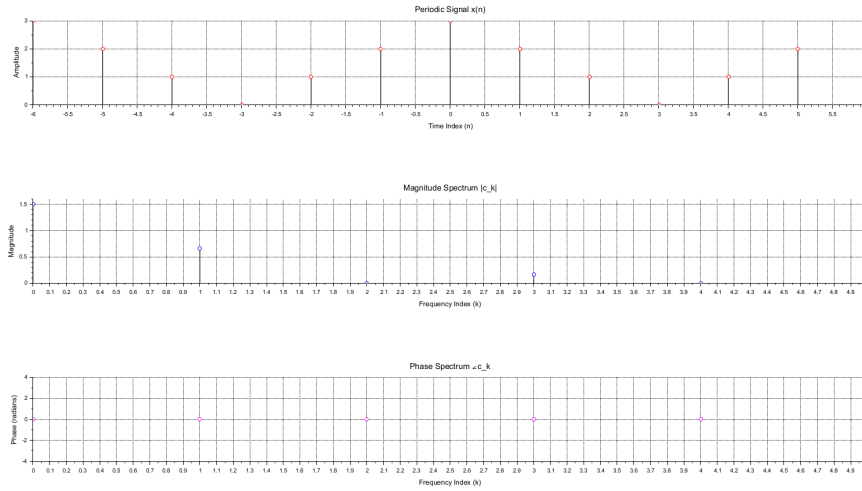


Figure 8: Time domain signal $x(n)$, Magnitude Spectrum $|c_k|$, and Phase Spectrum $\angle c_k$.
For $N = 6$, the coefficients are:

- $c_0 = \frac{1}{6}(3 + 2 + 1 + 0 + 1 + 2) = 1.5$
- $c_1 = c_5^* = 0.667$
- $c_2 = c_4^* = 0$
- $c_3 = \frac{1}{6}(3 - 2 + 1 - 0 + 1 - 2) = 0.167$

Since all c_k are real and non-negative, the **Phase Spectrum** $\angle c_k = 0$ for all k .

(b) Verification of Parseval's Relation

Parseval's relation states that $P_{time} = P_{freq}$.

Time Domain Power:

$$P_{time} = \frac{1}{6} \sum_{n=0}^5 |x(n)|^2 = \frac{1}{6}(3^2 + 2^2 + 1^2 + 0^2 + 1^2 + 2^2) = \frac{19}{6} \approx 3.1667$$

Frequency Domain Power:

$$P_{freq} = \sum_{k=0}^5 |c_k|^2 = (1.5)^2 + (0.667)^2 + (0)^2 + (0.167)^2 + (0)^2 + (0.667)^2$$

$$P_{freq} = 2.25 + 0.444 + 0 + 0.027 + 0 + 0.444 = \frac{114}{36} = 3.1667$$

As $P_{time} = P_{freq}$, Parseval's relation is verified.

4.5 Exercise 4.5

Solution

(a)

$$c_k = \frac{1}{8} \sum_{n=0}^7 x(n) e^{-j\pi k n/4}$$

$$x(n) = \left\{ 1, \frac{1}{\sqrt{2}}, \frac{2+3\sqrt{2}}{4\sqrt{2}}, 1, \frac{2-3\sqrt{2}}{4\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{2-3\sqrt{2}}{4\sqrt{2}}, 1, \frac{2+3\sqrt{2}}{4\sqrt{2}} \right\}$$

Hence, $c_0 = 2$, $c_1 = c_7 = 1$, $c_2 = c_6 = \frac{1}{\sqrt{2}}$, $c_3 = c_5 = \frac{1}{4}$, $c_4 = 0$

(b)

$$P = \sum_{i=0}^7 |c_i|^2$$

$$= 4 + 1 + 1 + \frac{1}{4} + \frac{1}{4} + \frac{1}{16} + \frac{1}{16} = \frac{53}{8}$$

4.6 Exercise 4.9

Solution

(a)

$$X(w) = \sum_{n=-\infty}^{\infty} x(n) e^{-jwn}$$

$$= \sum_{n=0}^5 e^{-jwn}$$

$$= \frac{1 - e^{-j6w}}{1 - e^{-jw}}$$

(b)

$$X(w) = \sum_{n=-\infty}^0 2^n e^{-jwn}$$

$$= \sum_{m=0}^{\infty} \left(\frac{e^{jw}}{2} \right)^m$$

$$= \frac{2}{2 - e^{jw}}$$

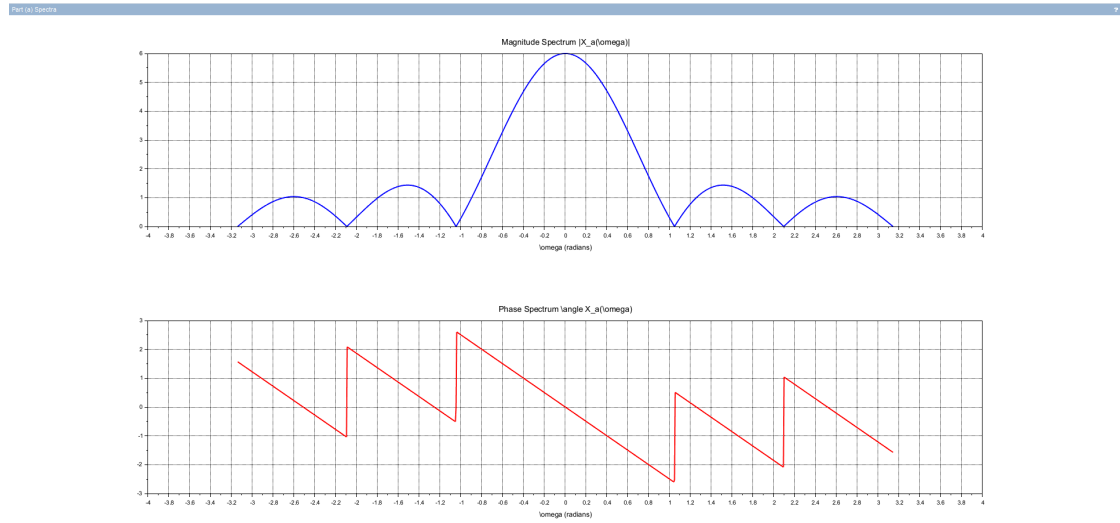


Figure 9: Caption

(c)

$$\begin{aligned}
 X(w) &= \sum_{n=-4}^{\infty} \left(\frac{1}{4}\right)^n e^{-jwn} \\
 &= \left(\sum_{m=0}^{\infty} \left(\frac{1}{4}\right)^m e^{-jmw}\right) 4^4 e^{j4w} \\
 &= \frac{4^4 e^{j4w}}{1 - \frac{1}{4}e^{-jw}}
 \end{aligned}$$

(d)

$$\begin{aligned}
 X(w) &= \sum_{n=0}^{\infty} \alpha^n \left(\frac{e^{jw_0 n} - e^{-jw_0 n}}{2j}\right) e^{-jwn} \\
 &= \frac{1}{2j} \sum_{n=0}^{\infty} \left(\alpha e^{-j(w-w_0)}\right)^n - \frac{1}{2j} \sum_{n=0}^{\infty} \left(\alpha e^{-j(w+w_0)}\right)^n \\
 &= \frac{1}{2j} \left[\frac{1}{1 - \alpha e^{-j(w-w_0)}} - \frac{1}{1 - \alpha e^{-j(w+w_0)}} \right] \\
 &= \frac{\alpha \sin(w_0) e^{-jw}}{1 - 2\alpha \cos(w_0) e^{-jw} + \alpha^2 e^{-j2w}}
 \end{aligned}$$

(e)

Note that:

$$\sum_{n=-\infty}^{\infty} x(n) = \sum_{n=-\infty}^{\infty} a^n \sin(w_0 n)$$

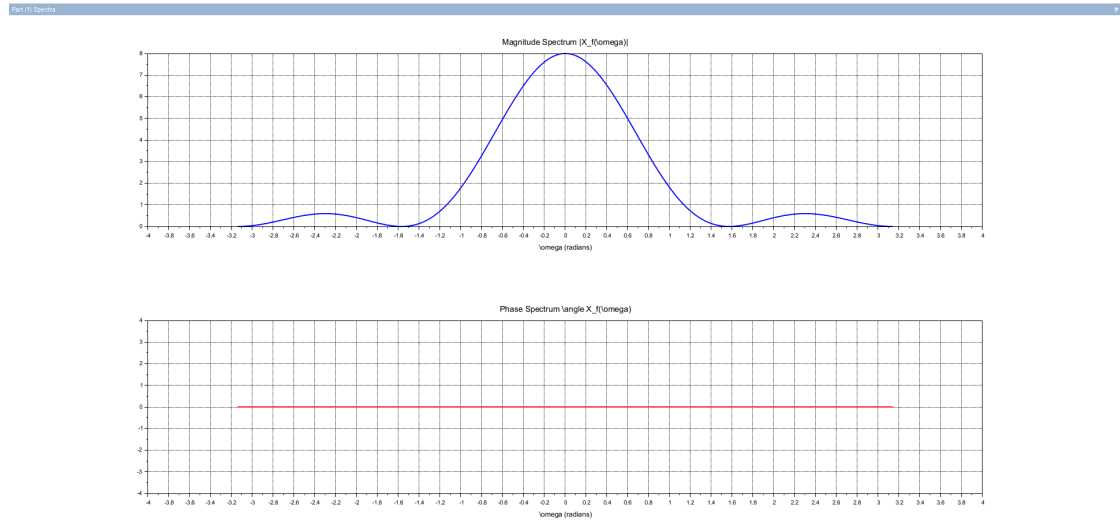


Figure 10: *Caption*

Suppose that $w_0 = \frac{\pi}{2}$, so that $\sin(w_0 n) = 1$

$$\sum_{n=-\infty}^{\infty} \alpha^n = \sum_{n=-\infty}^{\infty} |x(n)| \rightarrow \infty$$

Therefore, the Fourier transform does not exist.

(f)

$$\begin{aligned} X(w) &= \sum_{n=-4}^4 x(n)e^{-jwn} \\ &= \sum_{n=-4}^4 e^{-j12n} e^{-jwn} \\ &= \frac{2e^{j4w}}{1 - e^{-jw}} - \frac{1}{2} [-4e^{j4w} + 4e^{-j4w} - 3e^{j3w} + e^{-j3w} - 2e^{j2w} + 2e^{-j2w} - e^{jw} + e^{-jw}] \\ &= \frac{2e^{j4w}}{1 - e^{-jw}} + j [4 \sin(4w) + 3 \sin(3w) + 2 \sin(2w) + \sin(w)] \end{aligned}$$

(g)

$$\begin{aligned} X(w) &= \sum_{n=-\infty}^{\infty} x(n)e^{-jwn} \\ &= -2j (e^{j2w} - e^{-jw} + e^{jw} + 2e^{-j2w}) \\ &= -2j (2 \sin(2w) + \sin(w)) \end{aligned}$$

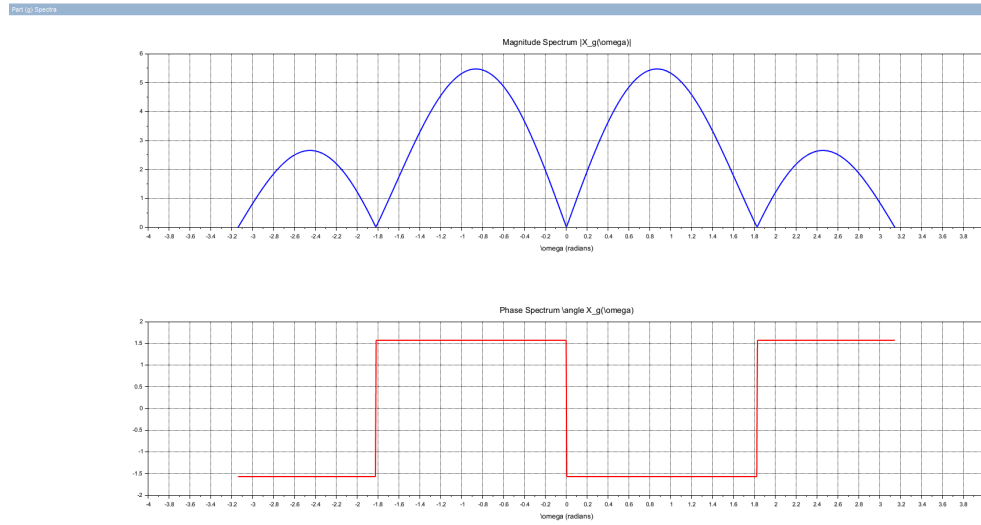


Figure 11: *Caption*

(h)

$$\begin{aligned}
 X(w) &= \sum_{n=-M}^M x(n)e^{-jwn} \\
 &= A \sum_{n=-M}^M (2M+1-n)e^{-jwn} \\
 &= A(2M+1) + A \sum_{k=1}^M (2M+1-k)(e^{-jwk} + e^{jwk}) \\
 &= A(2M+1) + 2A \sum_{k=1}^M (2M+1-k) \cos(wk)
 \end{aligned}$$

4.7 Exercise 4.10

Solution

(a)

$$\begin{aligned}
 x(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(w)e^{jwn} dw \\
 &= \frac{1}{2\pi} \int_{-\pi}^{-w_0} e^{jwn} dw + \frac{1}{2\pi} \int_{w_0}^{\pi} e^{jwn} dw \\
 x(0) &= \frac{1}{2\pi} (\pi - w_0) + \frac{1}{2\pi} (\pi - w_0) = \frac{\pi - w_0}{\pi}
 \end{aligned}$$

For $n \neq 0$,

$$\begin{aligned}\int_{-\pi}^{-w_0} e^{jwn} dw &= \left[\frac{1}{jn} e^{jwn} \right]_{-\pi}^{-w_0} = \frac{1}{jn} (e^{-jw_0 n} - e^{-j\pi n}) \\ \int_{w_0}^{\pi} e^{jwn} dw &= \left[\frac{1}{jn} e^{jwn} \right]_{w_0}^{\pi} = \frac{1}{jn} (e^{j\pi n} - e^{jw_0 n})\end{aligned}$$

Hence, $x(n) = -\frac{\sin(nw_0)}{n\pi}$, $n \neq 0$

(b)

$$\begin{aligned}X(w) &= \cos^2 w \\ &= \left(\frac{1}{2} e^{jw} + \frac{1}{2} e^{-jw} \right)^2 \\ &= \frac{1}{4} (e^{j2w} + 2 + e^{-j2w})\end{aligned}$$

$$\begin{aligned}x(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(w) e^{jwn} dw \\ &= \frac{1}{8\pi} [2\pi\delta(n+2) + 4\pi\delta(n) + 2\pi\delta(n-2)] \\ &= \frac{1}{4} [\delta(n+2) + 2\delta(n) + \delta(n-2)]\end{aligned}$$

(c)

$$\begin{aligned}x(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(w) e^{jwn} dw \\ &= \frac{1}{2\pi} \int_{w_0-\delta w/2}^{w_0+\delta w/2} e^{jwn} dw \\ &= \frac{\delta w}{2\pi} \frac{\sin(n\delta w/2)}{n\delta w/2} e^{jn w_0}\end{aligned}$$

(d)

$$\begin{aligned}x(n) &= \frac{1}{2\pi} \operatorname{Re} \left(\int_0^{\pi/8} 2e^{jwn} dw + \int_{\pi/8}^{3\pi/8} e^{jwn} dw + \int_{6\pi/8}^{7\pi/8} e^{jwn} dw + \int_{7\pi/8}^{\pi} 2e^{jwn} dw \right) \\ &= \frac{1}{\pi} \left(\int_0^{\pi/8} 2\cos(wn) dw + \int_{\pi/8}^{3\pi/8} \cos(wn) dw + \int_{6\pi/8}^{7\pi/8} \cos(wn) dw + \int_{7\pi/8}^{\pi} 2\cos(wn) dw \right) \\ &= \frac{1}{n\pi} \left(\sin \frac{7\pi n}{8} + \sin \frac{6\pi n}{8} - \sin \frac{3\pi n}{8} - \sin \frac{\pi n}{8} \right)\end{aligned}$$

4.8 Exercise 4.11

Solution

$$\begin{aligned}x_e(n) &= \frac{x(n) + x(-n)}{2} \\&= \left\{ \frac{1}{2}, 0, 1, 2, 1, 0, \frac{1}{2} \right\}\end{aligned}$$

$$\begin{aligned}x_o(n) &= \frac{x(n) - x(-n)}{2} \\&= \left\{ \frac{1}{2}, 0, -2, 0, 2, 0, \frac{1}{2} \right\}\end{aligned}$$

Then,

$$\begin{aligned}X_R(w) &= \sum_{n=-3}^3 x_e(n) e^{-jwn} \\jX_I(w) &= \sum_{n=-3}^3 x_o(n) e^{-jwn}\end{aligned}$$

Now,

$$Y(w) = X_I(w) + X_R(w) e^{j2w}$$

Therefore,

$$\begin{aligned}y(n) &= \mathcal{F}^{-1}\{X_I(w)\} + \mathcal{F}^{-1}\{X_R(w) e^{j2w}\} \\&= -jx_o(n) + x_e(n+2) \\&= \left\{ \frac{1}{2}, 0, 1-j2, 2, 1+j2, 0, \frac{1}{2}-j2, 0, j2 \right\}\end{aligned}$$

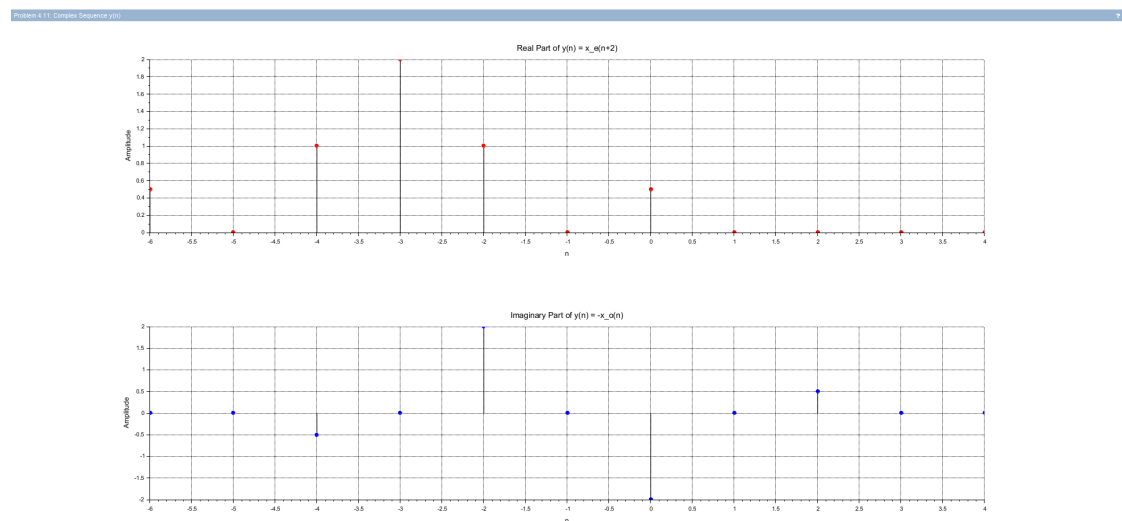


Figure 12: *Caption*



5 Reference

The full source code and this report is available in the GitHub repository:
<https://github.com/l1ttled1no/DigitalSignalProcessing-Lab>